ARGUS NEO: Joint Structural–Metrology Inference and Dual-Control Experiment Design for Adaptive Defect Localization

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Abstract*—Most data-driven nondestructive-evaluation systems passively classify a fixed measurement. This paper presents ARGUS NEO (Adaptive Recursive Guided Uncertainty Sensing—Next Experiment Optimization), a working research prototype that instead closes the loop between sensing and inference. NEO maintains a joint posterior over structural state and nuisance/metrology variables; chooses diagnostic or calibration actions through dual control; compares candidate-specific counterfactual observation distributions; trades Bayes risk reduction against motion, energy, redundancy, and safety constraints; combines analytical, physics, and learned fidelities with an online discrepancy model; detects out-of-distribution (OOD) observations; and stops, verifies, or abstains under explicit decision rules. The ARGUS-X assurance layer adds explicit healthy, damage-candidate, and unsupported states; accumulated channel reliability; environmental-envelope warnings; duplicate acquisition rejection; a persisted emergency-stop latch; and conservative human escalation. The system also includes bounded multi-family waveform co-design, receding-horizon lookahead, phone-camera probe guidance, an edge-node client, truth-sealed replay, and a hash-chained evidence ledger. In the earlier 30-case matched-model ARGUS baseline, normalized entropy ended at 0.419 versus 0.498 for random probing and 0.621 for a uniform grid; localization-error intervals crossed zero. New NEO smoke studies are reported as engineering diagnostics rather than efficacy claims: a two-case, five-measurement nine-policy matrix did not meet the stopping criterion or demonstrate compression, and four quick simulated calibration cases produced empirical coverage 1.0 but poor expected calibration error (0.715). In one controlled seed-17 model-mismatch demonstration, NEO reduced error from 157.5 mm to 14.6 mm while identifying OOD/caution, but this single case is not a population result. The implementation therefore establishes a reproducible platform and testable technical architecture, not validated inspection performance on blind physical defects.

***Index Terms*—active sensing, Bayesian experimental design, dual control, structural health monitoring, nondestructive evaluation, uncertainty quantification, digital twin, out-of-distribution detection, adaptive measurement**

I. INTRODUCTION

Nondestructive evaluation (NDE) is usually presented as a mapping from a measurement to a decision: acquire a waveform, compute features, and classify or localize damage. That formulation leaves the measurement geometry fixed. Yet the source position, receiver position, waveform, frequency band, amplitude, and duration determine which aspects of an internal structure are observable. When measurements have different costs, choosing the next experiment can be as important as interpreting the previous one.

TABLE I
PASSIVE DETECTION VERSUS ARGUS ACTIVE INTERROGATION

Property	Passive pipeline	ARGUS prototype
Measurement	Fixed plan	Chosen from current belief
State	Label or point	Normalized spatial posterior
Uncertainty	Often implicit	Entropy, covariance, local mass
History	Optional aggregation	Recursive Bayesian fusion
Output	Detect or locate	Map, next action, reason, stop state
Efficiency	Scan or fixed network	Information–cost tradeoff

ARGUS addresses the sequential question

$$e^{\star}_{t+1} = \arg\max_{e\in\mathcal{E}_t} \left[\mathcal{I}_t(e) - \mathcal{C}_t(e)\right], \tag{1}$$

where e is a physically executable experiment and $\mathcal{E}_t$ is a state-dependent candidate set. After observing response y_{t+1}, ARGUS performs a recursive Bayesian update, regenerates candidates from the new posterior, and repeats. This is related to Bayesian experimental design and active data selection [1], [2], but instantiated as a live, spatial, source–receiver interrogation loop.

The prototype targets an opaque rectangular panel with one dominant hidden anomaly (cavity, loose region, delamination, or dense inclusion). Its central distinction from a one-shot classifier is summarized in Table I. It is designed to run without hardware through a seeded physics-inspired simulator, while the same session engine accepts a microphone, validated WAV upload, or serial-connected probe.

The contributions of this implementation are:

1) a complete observe–infer–plan–act loop joining simulation or acquisition to a joint structural, material, sensor, and placement posterior;

2) a dual-control planner that switches between diagnostic and calibration actions and evaluates counterfactual observation distributions using risk, information, movement, energy, redundancy, and safety terms;
3) bounded co-design across nine waveform families, multiple source–receiver geometries and fidelity levels, with short receding-horizon lookahead;
4) robust likelihood tempering, online model-discrepancy correction, ensemble/conformal OOD indicators, calibrated-decision interfaces, and explicit verify/abstain states;
5) an auditable research stack comprising resumable jobs, paired benchmarks, ablations, fault injection, truth-sealed replay, hash-chained evidence export, phone-camera guidance, and a reconnecting edge node;
6) backward-compatible endpoints, 41 passing backend tests, three passing frontend utility tests, and a production frontend build that make every implemented mechanism independently inspectable.

The novelty claim is deliberately narrow. Guided-wave NDE, Bayesian sensor placement, active learning, dual control, and matched filtering all predate this work [3]–[5]. The research contribution is a particular executable composition: joint structural–metrology belief, diagnostic/calibration action selection, counterfactual physical co-design, discrepancy-aware fidelity and OOD gating, and cryptographically linked evidence. Legal patentability is not established by this manuscript and requires a professional claim-specific prior-art search.

II. Related Work and Design Position

A. Guided-Wave and Vibration-Based Inspection

Elastic waves can interrogate plate-like structures over areas larger than a probe footprint, while damage modifies propagation through scattering, attenuation, mode conversion, and local resonance [3]. Practical guided-wave interpretation is complicated by dispersion, anisotropy, boundaries, coupling variability, and environmental drift. Open Guided Waves provides controlled CFRP measurements for comparative algorithm research [6], illustrating both the value and the complexity of measured wavefields.

ARGUS does not claim that its compact simulator is a high-fidelity Lamb-wave or finite-element model. It uses path length, propagation delay, attenuation, signed defect reflection, frequency-sensitive gain, and damped ringing to create an interpretable inverse problem that can exercise the complete adaptive loop. The real-device path is intentionally decoupled from this simulator.

B. Bayesian Design and Active Measurement

Bayesian experimental design selects an experiment according to expected utility under current uncertainty [1]; information-based active selection chooses data expected to be informative about hypotheses or predictions [2]. In structural health monitoring, Flynn and Todd formulated optimal actuator/sensor placement using Bayes risk and demonstrated active guided-wave examples [4]. Much of that literature optimizes a fixed network or configuration. ARGUS instead replans a single next, executable experiment after every observation and incorporates movement, excitation, and redundancy costs.

Exact expected information gain would marginalize over all possible future observations for every candidate. That calculation is expensive and only as reliable as the generative model. ARGUS therefore uses a bounded counterfactual-disagreement proxy. Throughout this paper, "expected information gain" means this implemented proxy, not an exact mutual-information calculation or a guarantee of global optimality.

C. Open Experimental Evidence

The LMSD 2021 dataset contains measured frequency-response functions (FRFs) from a $600 \times 600 \times 4$ mm cross-ply CFRP plate, a 12×12 coordinate grid, seven accelerometer/hammer nodes, a healthy baseline, and six added-mass scenarios [7]. It supplies real multistatic responses and known mass locations, but not cavities, delaminations, or adaptively selected trials. ARGUS uses it to test data ingestion and feature transfer, not to imply physical validation of every supported defect type.

III. Problem Formulation

Let the panel occupy normalized coordinates $\Omega = [0,1]^2$. Physical distance is evaluated after the transformation

$$\boldsymbol{p}_{\mathrm{m}} = \mathrm{diag}(W,H)\boldsymbol{p}, \tag{2}$$

where the default width and height are $W = 0.60$ m and $H = 0.40$ m. The latent defect state is

$$z = (x, y, r_x, r_y, q, T), \tag{3}$$

containing center, elliptical radii, severity $q \in [0,1]$, and type T. The current prototype explicitly infers only location (x,y) on an $G \times G$ grid; size, severity, and type are nuisance variables captured by simulator randomization and likelihood robustness.

An experiment is

$$e = (\boldsymbol{s}, \boldsymbol{r}, f_0, f_1, a, d, w), \tag{4}$$

where $\boldsymbol{s}$ and $\boldsymbol{r}$ are normalized source and receiver coordinates, f_0, f_1 are band endpoints, a is normalized amplitude, d is duration, and $w \in \{\text{impulse,sine,chirp}\}$. Given observation history $\mathcal{D}_t = \{(e_i, y_i)\}_{i=1}^{t}$, the belief grid is

$$P_t(j) = P(z_{xy} = j \mid \mathcal{D}_t), \qquad \sum_{j=1}^{G^2} P_t(j) = 1. \tag{5}$$

The initial belief is uniform. Each observation supplies a likelihood $L_{t+1}(j)$ and

$$P_{t+1}(j) = \frac{P_t(j)\max[L_{t+1}(j), \varepsilon]}{\sum_k P_t(k)\max[L_{t+1}(k), \varepsilon]}, \quad \varepsilon = 10^{-12}. \tag{6}$$

The normalized uncertainty is

$$\bar{H}(P) = -\frac{\sum_j P(j)\log_2(P(j)+\varepsilon)}{\log_2(G^2)} \in [0,1]. \tag{7}$$

IV. System Architecture

Figure 1 shows the implemented data flow. Ground truth exists only inside simulation sessions and remains server-side until an explicit reveal action. Both simulated and physical signals pass through the same analysis, persistence, and user-interface contracts.

The Python backend contains domain models, simulation, signal analysis, belief inference, planning, evaluation, device adapters, session orchestration, and SQLite persistence. A FastAPI layer exposes session, recommendation, experiment, upload, posterior, history, reveal, device, and benchmark endpoints. The TypeScript/React frontend visualizes the posterior heatmap, recommended geometry, signal, spectrum, power spectral density, spectrogram, belief history, benchmark curves, and camera overlay. A single launcher starts both processes locally.

V. Physics-Inspired Forward Model

A. Excitation

For $N = \lfloor d f_s \rceil$ samples at default sample rate $f_s = 16$ kHz, the excitation occupies the first

$$N_b = \lfloor f_s \min(10\,\mathrm{ms}, 0.22d) \rceil \tag{8}$$

samples. A Tukey window with shape parameter 0.45 suppresses burst discontinuities [8]. A sine uses $\sin(2\pi f_0 t)$; a chirp sweeps linearly from f_0 to f_1; and the impulse option uses a Ricker-like pulse

$$u(t) = a[1 - 2\phi(t)^2] \exp[-\phi(t)^2], \tag{9}$$

$$\phi(t) = \frac{\pi(t - 0.28 d_b)}{\max(f_c^{-1}, 0.22\,\mathrm{ms})}. \tag{10}$$

B. Direct and Scattered Responses

Let d_{sr} denote physical source–receiver distance and let d_{sz}, d_{zr} denote the two legs through the defect. The response is

$$y(t) = A_0 u(t - \tau_0) + A_z u(t - \tau_z) + r_z(t) + n(t) + b(t), \tag{11}$$

with fractional delays obtained by linear interpolation. For wave velocity c, attenuation α, and system delay t_0,

$$\tau_0 = t_0 + d_{sr}/c, \qquad A_0 = \frac{e^{-\alpha d_{sr}}}{1 + 4 d_{sr}}, \tag{12}$$

$$\tau_z = t_0 + (d_{sz} + d_{zr})/c. \tag{13}$$

The defect resonance and scattered gain are

$$f_z = f_{\mathrm{mat}}(1.05 - 0.32q), \tag{14}$$

$$R_f = 0.72 + 0.72 \exp\left[-\frac{1}{2}\left(\frac{f_c - f_z}{1250}\right)^2\right], \tag{15}$$

$$A_z = 0.43 \rho_T q \frac{\sqrt{r_x r_y}}{0.085} \frac{e^{-\alpha(d_{sz} + d_{zr})}}{1 + 5(d_{sz} + d_{zr})} R_f, \tag{16}$$

where $\rho_T = 1.00, 0.72, 0.86, -0.64$ for cavity, loose region, delamination, and dense inclusion, respectively. The negative dense-inclusion coefficient creates a phase inversion. Starting at τ_z,

TABLE II
ARGUS Response Features

Group	Features
Amplitude	RMS, peak, crest factor
Temporal	zero-crossing rate, envelope-peak time, $1/e$ decay time
Spectral shape	centroid, bandwidth, 85% rolloff, dominant frequency, normalized entropy
Spectral allocation	low-, mid-, and high-band energy fractions
Quality	RMS-to-robust-noise SNR in dB

$$r_z(t) = 0.11 A_z a \sin[2\pi f_r (t - \tau_z)] e^{-\beta(t - \tau_z)}, \tag{17}$$

where $f_r = f_z$ except for dense inclusions, for which $f_r = 1.12 f_z$. Noise $n(t)$ is Gaussian; $b(t)$ is a cumulative low-frequency random drift. The default material values are $c = 180$ m/s, $\alpha = 1.65$ m^{-1}, $f_{\mathrm{mat}} = 3100$ Hz, $\beta = 95$ s^{-1}, $t_0 = 0.8$ ms, and noise standard deviation 0.007.

This model is intentionally identifiable and fast enough for interactive candidate evaluation. It omits dispersion branches, anisotropic stiffness, boundary-mode conversion, transducer electromechanics, and spatially varying coupling. Consequently, simulation accuracy must not be confused with field accuracy.

VI. Signal Interpretation and Recursive Belief

A. Preprocessing and Features

The pipeline rejects nonfinite or shorter-than-eight-sample signals, removes the mean, and optionally applies a fourth-order zero-phase Butterworth bandpass from 250 Hz to $\min(7000, 0.96 f_s/2)$ Hz. Peak normalization is optional and disabled for feature extraction so amplitude remains informative. A Hann-windowed real FFT produces a unit-sum spectral energy distribution. Welch PSD, a short-time spectrogram, and a Hilbert envelope support inspection [9], [10].

Table II lists the 15 learned forward-response targets. The runtime inspector additionally returns a robust noise estimate. Low-, mid-, and high-band fractions span 250–1500, 1500–3500, and 3500–7000 Hz (clipped by Nyquist). Spectral entropy is normalized by the log of the number of FFT bins. Decay time is the first post-envelope-peak sample below $1/e$ of the peak. The noise scale is 1.4826 times the median absolute deviation (MAD) in the first 10% of the record.

B. Matched-Filter Spatial Likelihood

The inference engine subtracts the predicted healthy direct response, giving residual $v = y - \hat{y}_0$. It correlates v with the excitation using an FFT implementation, retains nonnegative lags, takes absolute magnitude, and applies a seven-sample maximum filter. Correlation is the classical matched-filter operation for a known waveform in noise [5].

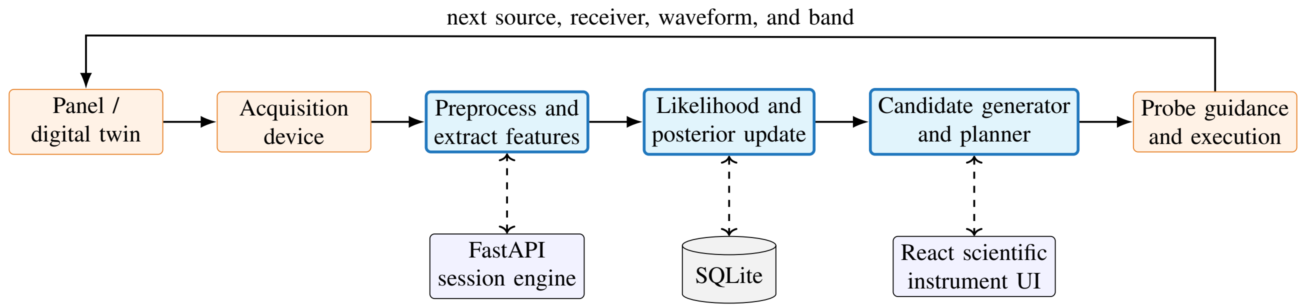

Fig. 1. ARGUS closed-loop architecture. The planner changes the next physical experiment; it does not merely classify a fixed waveform.

For every grid-cell center j, the model predicts the bistatic delay

$$\hat{\tau}_j = t_0 + \frac{d(\boldsymbol{s}, \boldsymbol{z}_j) + d(\boldsymbol{z}_j, \boldsymbol{r})}{c}. \tag{18}$$

The correlation value at the nearest nonnegative lag is assigned to that cell, smoothed spatially by a Gaussian with $\sigma = 0.55$ cells, and robustly scaled between its 8th and 98th percentiles to $\tilde{C}_j \in [0, 1]$. The likelihood is

$$L(j) \propto \exp(T_{\text{eff}} \tilde{C}_j), \tag{19}$$

$$T_{\text{eff}} = 4.8 \,\text{clip}\left(\frac{\text{SNR} - 0.75}{1.8}, 0.30, 1 \right), \tag{20}$$

where SNR is residual RMS divided by residual MAD scale. Scaling L to have mean one makes the Bayesian update numerically convenient without altering Eq. (6).

C. Estimation, Confidence, and Termination

ARGUS reports both the maximum-a-posteriori (MAP) grid center and posterior mean. The 2×2 covariance is the posterior-weighted covariance of cell coordinates and yields a confidence ellipse. Let m_{local} be posterior mass in a square neighborhood of radius $\max(1, \lfloor G/16 \rfloor)$ around the MAP cell. The displayed confidence is the bounded heuristic

$$Q = \text{clip}\{0.55 m_{\text{local}} + 0.45[1 - \bar{H}(P)], 0, 1\}. \tag{21}$$

The default controller stops when $Q \geq 0.72$, $\bar{H}(P) \leq 0.38$, or 12 experiments have been executed. A benchmark may impose a different budget; the reported study used ten.

VII. Adaptive Physical Experiment Planner

A. Candidate Construction

The generator mixes eight perimeter locations with the eight highest-posterior cells as candidate sources. Receivers are the four perimeter corners plus the center. It cycles through bands 1.2–3.0, 2.2–4.4, and 3.4–6.2 kHz, with chirp, impulse, and chirp waveforms, amplitudes 0.42, 0.48, or 0.54, and 0.12 s duration. Near-coincident source–receiver pairs are redirected. Duplicate parameter tuples are removed until the default pool contains 48 candidates.

B. Counterfactual Hypothesis Disagreement

Let $\mathcal{K}$ contain the top $K = 24$ grid hypotheses, with represented posterior mass $M = \sum_{i \in \mathcal{K}} P(i)$ and normalized weights $w_i = P(i)/M$. For candidate e, the fast forward signature is

$$\boldsymbol{g}_i(e) = \left[\tau_i, \log(A_i + 10^{-8}), \sin\varphi_i, \cos\varphi_i\right], \quad \varphi_i = 2\pi f_c \tau_i, \tag{22}$$

where $A_i = 0.65 e^{-\alpha d_i}/(1 + 5d_i)$. Delay and log-gain dimensions are divided by 0.18 ms and 0.55, respectively, while sine/cosine encode circular phase without wrap discontinuities. With scaled signature $\tilde{\boldsymbol{g}}$, pairwise distinguishability and disagreement are

$$D_{ij}(e) = 1 - \exp\left[-\tfrac{1}{2}\|\tilde{\boldsymbol{g}}_i(e) - \tilde{\boldsymbol{g}}_j(e)\|_2^2\right], \tag{23}$$

$$D(e) = \sum_{i,j \in \mathcal{K}} w_i w_j D_{ij}(e). \tag{24}$$

The bounded information-gain proxy is

$$I(e) = H(P) M D(e), \tag{25}$$

in bits. It grows with current uncertainty, posterior mass represented by the leading hypotheses, and predicted separability. It is not the expected value of Eq. (7) under a calibrated observation distribution.

Candidate coverage is

$$U(e) = \sum_{i \in \mathcal{K}} w_i \exp[-2.2 \min(d_{si}, d_{ir})]. \tag{26}$$

The implemented physical cost is

$$C(e) = 0.52 a^2 \frac{d}{0.12} + 0.48 \frac{\|\boldsymbol{s} - \boldsymbol{s}_t\|_2}{\sqrt{2}}, \tag{27}$$

with zero motion term for the first experiment. Redundancy is the closest match to any previous experiment:

$$R(e) = \max_{h<t} \exp[-6\|\boldsymbol{s} - \boldsymbol{s}_h\|_2] \exp[-2|f_c - f_{c,h}|/3000]. \tag{28}$$

The final score is

$$S(e) = I(e) + 0.35 D(e) + 0.22 U(e) - 0.16 C(e) - 0.22 R(e). \tag{29}$$

The highest-scoring candidate is selected. The interface exposes all terms for the top five candidates, preventing the planner from becoming an opaque recommendation engine.

Algorithm 1: ARGUS closed-loop localization

Input: panel model, stopping thresholds, budget B.
Set $P_0(j) \leftarrow 1/G^2$ and $\mathcal{D} \leftarrow \emptyset$.
for $t = 0, \dots, B-1$ **do**
  Form sources from perimeter and leading cells.
  Generate and deduplicate candidate experiments.
  Retain top-K posterior hypotheses.
  for each candidate e **do**
    Predict $\{\boldsymbol{g}_i(e)\}_{i=1}^{K}$.
    Compute $I(e), D(e), U(e), C(e), R(e)$.
    $S(e) \leftarrow I + .35D + .22U - .16C - .22R$.
  end for
  Set $e_{t+1} \leftarrow \arg\max_e S(e)$ and execute it.
  Acquire y_{t+1} and construct L_{t+1}.
  $P_{t+1} \leftarrow \text{normalize}(P_t \odot L_{t+1})$.
  Persist signal, features, scores, and posterior.
  break if confidence or entropy threshold holds.
end for
Return: MAP, mean, covariance, confidence, history.

C. Programmatic Explanation

The selected recommendation receives a deterministic rationale. High disagreement produces a competing-regions explanation; broad entropy produces a coverage explanation; otherwise the system describes local ambiguity resolution. If repetition is below 0.25 after prior measurements, the explanation notes that the geometry and frequency avoid recent experiments. No language model is required, and every statement traces to planner state.

VIII. ARGUS NEO Inference and Control Extension

A. Joint Structural–Metrology State

The original spatial grid is retained as the structural marginal so earlier clients remain compatible, but NEO augments it with nuisance and metrology variables. The sequential state is

$$p_t(\boldsymbol{z}, \boldsymbol{\eta}) = p_t(x, y, r, q, k, \boldsymbol{\eta} \mid \mathcal{D}_t), \tag{30}$$

where (x, y) is location, r is effective size, q severity, k defect family, and $\boldsymbol{\eta}$ contains wave velocity, attenuation, timing offset, noise scale, gain, source/receiver coupling, and source/receiver pose error. The implemented compact approximation stores a normalized structural grid plus bounded Gaussian nuisance marginals and particles. It is therefore a factorized engineering approximation to Eq. (30), not an exact high-dimensional posterior.

Each valid observation updates the structural likelihood and the nuisance sufficient statistics. Estimated velocity and timing are synchronized back into inference so calibration actions alter subsequent predictions. Signal quality produces evidence weight $0 \le w_t \le 1$ from SNR, clipping, silence, acceleration deviation, and visual pose error. The recursive update is tempered as

$$p_{t+1}(\boldsymbol{z}) \propto p_t(\boldsymbol{z}) L(y_{t+1} \mid \boldsymbol{z}, e_{t+1}, \hat{\boldsymbol{\eta}}_t)^{w_t/T_t}, \tag{31}$$

where T_t also increases under mismatch. Silence or unusable acquisition cannot sharpen belief because its evidence weight approaches zero.

B. Dual Control: Diagnose or Calibrate

An action may interrogate the object or reduce uncertainty in the measuring process itself. Let U_s denote structural uncertainty, U_m combined metrology uncertainty, ρ model trust, and o the OOD score. NEO compares

$$V_{\text{cal}} = [U_m(0.70 + 0.30(1-\rho)) + 0.35o]\, d_n, \tag{32}$$

$$V_{\text{diag}} = \hat{I}(e) U_s \rho, \tag{33}$$

with diminishing-return factor $d_n = (1+0.65n_{\text{cal}})^{-1}$. Calibration is selected when $V_{\text{cal}} - V_{\text{diag}} > 0.05$, or when an explicit threshold/OOD rule requires it. The dominant nuisance term maps to an executable direct-path, coupling-repeat, phone-pose, microphone-level, or frequency-sweep action. Otherwise NEO selects diagnostic, exploration, or pre-stop verification. This is a transparent heuristic dual controller whose components and switching margin are returned by the API.

C. Counterfactual Observation Distributions and Bayes Risk

For the leading hypotheses h_i, a candidate action e produces compact predicted distributions $q(y \mid h_i, e, \boldsymbol{\eta})$ rather than only a point score. The implementation combines standardized mean separation and predictive variance into average and worst-pair separation. Decision risk is explicitly parameterized:

$$R_t = 12 p_{\text{miss}} + 3 p_{\text{false alarm}}, \quad p_{\text{miss}} = \text{clip}(1 - c_t + 0.45 o_t), \tag{34}$$

and expected risk reduction is bounded by predicted information, separation, and model trust. The default multiobjective utility is

$$\begin{aligned} J(e) = {} & I(e) + 0.35D(e) + 0.55\Delta R(e) \\ & + 0.65V_{\text{cal}}(e) + 0.30\rho + \beta U(e) \\ & - 0.16C(e) - 0.22R(e) - 0.08T(e) - 0.08E(e), \end{aligned} \tag{35}$$

where β depends on exploration state. Alternative objectives expose information gain, Bayes risk, worst-case ambiguity, and information per physical cost. False-negative, false-positive, escalation, abstention, movement, time, and excitation losses are configuration values; they are research utilities, not certified maintenance economics.

D. Joint Geometry–Waveform and Horizon Search

Source position, receiver position, band, amplitude, duration, and waveform are optimized under hard limits. The candidate library supports impulse, sine, chirp, tone burst, Ricker, multisine, phase-coded, complementary-coded, and spectrally notched excitation. Every generated waveform is peak-normalized after modulation/coding, and actions exceeding frequency, amplitude, duration, minimum-spacing, no-go-region, or device-availability constraints are rejected with machine-readable reasons. A cached analytical level and

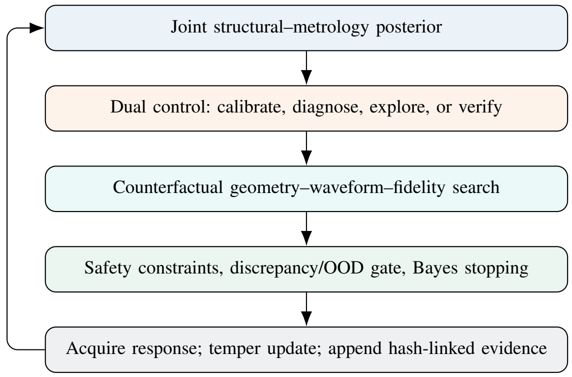

Fig. 2. ARGUS NEO closes inference, instrument calibration, experiment design, safety, and provenance in one recursive loop.

physics-signature level are selected according to uncertainty, ambiguity, frequency, and trust. A beam-limited receding-horizon reranker evaluates one or two steps depending on the execution profile; this is short lookahead, not a global policy solution.

E. Discrepancy, OOD, and Decision Gating

An online ridge discrepancy model learns residual feature correction between predicted and observed responses and converts residual error to model trust $\rho \in [0, 1]$. Robust residual distance, an ensemble disagreement score, and a conformal tail score are combined for OOD monitoring; the conformal component remains withheld until ten reference scores exist. Reported decision confidence is capped by OOD evidence. The stopping controller considers confidence, normalized entropy, credible-region area, Bayes risk, expected value of another measurement, budget, and model trust. It may continue, verify, escalate to a reference method, or abstain. These mechanisms reduce unsupported certainty but do not guarantee statistical calibration under physical domain shift.

F. Evidence, Replay, and Distributed Operation

Every state-changing event can be recorded as canonical JSON with timestamp, prior hash, and SHA-256 entry hash. Exported ZIP bundles contain a manifest, session state, ledger, responses, planner explanations, configuration, versions, and checksums; import verifies integrity before accepting the bundle. Replay readers support NPZ, CSV, and WAV response banks, while hidden truth remains sealed until the policy has stopped. SQLite-backed research jobs provide progress, cancellation, and interrupted-job recovery for counterfactual-bank generation, calibration, benchmarks, ablations, and surrogate studies. A phone progressive web application supplies camera overlay, inertial-quality metadata, and microphone acquisition; a reconnecting laptop edge client can register, heartbeat, poll, and return microphone measurements over a private LAN. These paths remain operator-supervised research tooling.

G. ARGUS-X Runtime Assurance and Failure Disposition

The extended system reports a normalized screening distribution over healthy/no-detectable-damage, known-damage-candidate, and unknown/unsupported states. Let $p_d = \sigma(\ell_d)$ summarize accepted residual-SNR evidence, o the OOD score, and ρ model trust; unsupported mass uses $\max(o, 1 - \rho)$ before renormalization. This transparent rule is not a calibrated correctness or POD claim. Separated posterior modes expose zero, one, or multiple candidate regions; interacting multi-scatterer inference remains open.

Each sensor channel maintains beta-binomial reliability

$$\hat{r}_j = \frac{\alpha_j}{\alpha_j + \beta_j}, \tag{36}$$

where accepted observations add quality-weighted evidence and rejections add failure evidence. Degraded channels cap confidence and request redundancy; unreliable channels force abstention. Session-baseline environment metadata provides drift warnings. Duplicate payloads/identifiers, invalid units or rates, non-finite samples, timestamp skew, clipping, silence, and infeasible actions are rejected before inference.

A persisted emergency latch blocks every acquisition path until a human supplies an acknowledgment and reason; both transitions are audited. Actions never include “structure safe”: ARGUS may continue, reacquire/repair, mark a region for reference inspection, require a human, or report no damage evidence only under tested conditions.

The 1,331-item ARGUS-X inventory is converted deterministically into a register: 277 narrow controls are tested, 588 partially mitigated, 244 require physical validation, 45 require literature/legal review, and 177 are bounded. These are dispositions, not 1,331 solved claims.

IX. Learned Forward Surrogate and Data

A. Domain-Randomized Synthetic Set

The optional learned component approximates expected response features, not the final posterior. Each row contains 24 inputs describing defect state, panel/material nuisance variables, and experiment parameters, and 15 outputs from Table II. Three thousand examples were generated with locations in $[0.08, 0.92]^2$, radii 0.04–0.14, severity 0.35–0.96, all four defect types, panel sizes 0.45–0.85 by 0.30–0.60 m, velocity 145–245 m/s, attenuation 1.0–2.4 m^{-1}, resonance 2.5–4.2 kHz, damping 70–135 s^{-1}, noise 0.003–0.014, and randomized probes, bands, amplitudes, and waveforms.

The PyTorch [11] network is

$$24 \to 96 \to 96 \to 15, \tag{37}$$

with SiLU activations and LayerNorm after the first hidden layer. Inputs and targets are standardized on training statistics. Training uses Smooth L1 loss, AdamW at learning rate 1.8×10^{-3} and weight decay 10^{-4}, batch size 128, a seeded 70/15/15 split, and early stopping. The physics predictor remains the default planner model because it is interpretable and the surrogate has uneven feature accuracy.

B. Measured LMSD Adapter

The downloader queries the official Zenodo record, downloads declared NPZ files, and verifies each published MD5 checksum. The adapter converts each complex 0–1600 Hz FRF to a 0.12 s impulse response, normalizes known node coordinates, encodes the added mass as a dense inclusion, and extracts the same feature contract. Six scenarios times seven sources times seven receivers yield 294 channels. To prevent leakage, splitting is by entire damage scenario: 196 training, 49 validation, and 49 test channels. These channels are correlated views of six physical damage configurations, not 294 independent defects.

X. Physical and Software Realization

A. Acquisition Abstraction

All devices implement connect, disconnect, acquire, and status operations. `SimulationDevice` executes Eq. (11); `MicrophoneDevice` records through the local audio stack; `UploadedSignalDevice` accepts WAV files; and `SerialProbeDevice` performs a line-framed transaction with an ESP32. Missing audio or serial hardware is reported without disabling simulation.

Uploads are limited to 10 MB, checked for allowed media types, decoded in memory, downmixed to mono, resampled by polyphase filtering, checked for finite values, and padded or truncated to the configured duration. The API never accepts arbitrary filesystem paths. Simulation ground truth is omitted from ordinary session payloads until reveal.

B. Calibration and Coordinates

The startup routine executes three chirps across distinct geometries and estimates noise standard deviation, effective propagation velocity, baseline RMS, and resonance. During a session, the dual controller can additionally request direct-path timing, repeated coupling, phone-pose, microphone-level, or frequency-sweep calibration. Values update the nuisance posterior and are persisted in the panel's calibration profile. Normalized coordinates keep the planner independent of physical dimensions, while the UI presents millimeters using Eq. (2).

A camera-guidance mode asks the operator to click the four panel corners in top-left, top-right, bottom-right, bottom-left order. Eight homography parameters are solved from the four point correspondences [12]. The browser overlays source, receiver, propagation path, posterior field, suspected region, and a 90% covariance ellipse. Visual-position error and inertial deviation are sent as quality metadata and reduce evidence weight; this remains approximate operator guidance, not certified optical metrology.

C. ESP32 Probe

The reference firmware uses GPIO25 for an exciter-driver control, GPIO34 for a conditioned analog sensor, GPIO2 for status, and 115200 baud serial. It accepts `PING` and an `EXPERIMENT` command containing frequency, amplitude, duration, waveform, and sample rate, then returns `BEGIN`, framed `DATA`, and `END`. An exciter must be driven through a MOSFET/transistor or amplifier; it must not be powered directly from a GPIO. A piezo input requires current limiting, clamping, biasing, and conditioning so the ADC never sees negative voltage or an overvoltage transient.

D. Persistence and Reproducibility

Additive SQLite migrations preserve legacy sessions while storing joint state, probes, events, research jobs, model records, and hash-linked ledger entries in addition to session configuration, calibration, compressed responses, features, candidate scores, and posterior history. NumPy/SciPy provide vectorized numerics and signal processing [13], [14]. Fixed seeds cover defect generation, observation noise, dataset splitting, training, and paired policy comparison. The API schema and saved JSON/CSV/SVG outputs allow an independent run to inspect reported metrics.

XI. Evaluation Protocol

The evaluation addresses four questions:

- **RQ1:** Does active planning concentrate spatial belief faster than random or uniform probing?
- **RQ2:** Does lower uncertainty translate into lower localization error under the same budget?
- **RQ3:** Can a small learned surrogate reproduce domain-randomized signal features?
- **RQ4:** Do those learned features transfer across held-out physical damage scenarios?

For RQ1–RQ2, 30 medium-preset defects were generated with seeds 100–129. Each of three policies operated on the identical hidden defect for a maximum of ten measurements. *Random* sampled continuous source and receiver positions and a frequency band. *Uniform grid* followed a fixed geometry sequence. *ARGUS* used Eq. (29). Policies were compared using mean and median localization error, normalized final entropy, entropy reduction, entropy area under the measurement curve (AUC), physical cost, experiment count, and success within 10, 15, 20, and 30 mm. Paired bootstrap 95% intervals were computed over the shared cases. No assertion required error to improve monotonically.

For RQ3, the 3000-example synthetic model used seed 23 and independent random splitting. For RQ4, the LMSD model used the same architecture and seed but held out complete scenarios. Smooth L1 values across these two data regimes are not directly comparable because their target statistics and split difficulty differ.

XII. Results

A. Planner Benchmark

Table III reports the saved benchmark, not hand-entered target values. ARGUS achieved the lowest mean error, final entropy, entropy AUC, and mean experiment count. At 15 mm it localized 73.3% of cases, versus 56.7% for random and 60.0% for the grid. At 20 mm all methods reached 93.3%; therefore the strongest separation occurs at the tighter 15 mm criterion.

TABLE III
PAIRED SIMULATION BENCHMARK: 30 MEDIUM-PRESET DEFECTS, TEN-EXPERIMENT BUDGET

Policy	Mean error (mm)	Median error (mm)	Mean exp.	Final $\bar{H}$	Entropy red.	Entropy AUC	Cost	≤ 15 mm (%)	≤ 20 mm (%)
Random	13.38	14.44	10.00	0.498	0.502	0.748	2.665	56.7	93.3
Uniform grid	13.26	13.84	10.00	0.621	0.379	0.818	**2.481**	60.0	93.3
ARGUS	**12.33**	**13.28**	**9.83**	**0.419**	**0.581**	**0.740**	2.588	**73.3**	93.3

TABLE IV
PAIRED ARGUS ADVANTAGES (95% BOOTSTRAP CI)

Comparison	Mean error advantage (mm)	Mean entropy advantage	Win rate (error / entropy)
vs. random	1.048 $[-0.667, 2.812]$	0.079 $[0.063, 0.095]$	0.30 / 0.93
vs. grid	0.928 $[-1.021, 2.757]$	0.202 $[0.180, 0.224]$	0.30 / 1.00

Relative to random probing, ARGUS lowered final entropy by 15.9%, lowered mean error by 7.8%, and used 2.9% less modeled cost. Relative to the grid, it lowered final entropy by 32.5% and mean error by 7.0%, but used 4.3% more modeled cost. The result illustrates Eq. (1): information efficiency and physical cost are related but distinct objectives.

Figure 3 shows mean trajectories. ARGUS does not dominate localization error at every intermediate step; its mean error is especially high after the second measurement and briefly rises again after the sixth. This is permissible in a noisy recursive estimator and reinforces why entropy and error must both be reported.

Paired results in Table IV qualify the aggregate means. ARGUS reduced final entropy in 28/30 cases versus random and 30/30 versus grid, with both confidence intervals above zero. In contrast, both mean-error intervals include zero and strict error win rate is 30%. Thus RQ1 is supported in this simulator, while RQ2 remains suggestive rather than statistically established. Tied MAP cells and a few large improvements explain why mean error can improve despite a low strict-win fraction.

B. NEO Engineering Diagnostics

Table V reports newly executed, seeded checks of the expanded implementation. They are deliberately separated from the 30-case baseline because the quick NEO matrix contains only two cases and a five-measurement budget. None of nine policies met the joint stopping criterion, all used five measurements, and therefore observed inspection compression was $1.0\times$. Full NEO mean error was 21.70 mm and success within the configured radius was 0.50; uniform, Bayes-risk-only, dual-control-only, and one-step horizon variants were 15.95 mm and 1.00 in these two cases. This is a pipeline smoke test and supplies no basis for a superiority claim.

The controlled mismatch demonstration altered effective velocity by 2% and attenuation by 4% under the same hidden truth, seed, and budget. NEO used eight diagnostic, three calibration, and six verification actions, ended at 14.65 mm error, model trust 0.517, and a caution state. The deliberately naive controller used 15 diagnostic and no calibration actions, ended at 157.54 mm with confidence 0.735, and triggered the false-confidence diagnostic. This one paired case illustrates the intended failure-handling mechanism; it is neither an estimated effect size nor evidence of physical generalization. In four calibration smoke cases, nominal 50–95% region coverages were all 1.0 but expected calibration error was 0.715, showing why tiny-sample apparent coverage must not be advertised as calibration.

TABLE V
OBSERVED ARGUS NEO DIAGNOSTIC RUNS

Run	Cases	Key error	Key uncertainty	Interpretation
Nine-policy quick matrix	2	NEO 21.70 mm	$\bar{H} = 0.909$	No criterion/compression
Quick calibration	4	ECE 0.715	Coverage 1.0	Too small; poorly calibrated
Mismatch, seed 17	1 paired	14.65 vs. 157.54 mm	NEO caution/OOD	Mechanism demo only
Active surrogate	180	MAE 94.10→89.01	5.41% decrease	Synthetic oracle only

TABLE VI
LEARNED FORWARD-RESPONSE EVALUATION

Data	Samples	Test Smooth L1	Std. MAE	Mean R^2	Split
Synthetic	3000	0.2245	0.5014	0.4092	seeded 70/15/15
LMSD measured	294	0.1902	0.4485	0.0010	held-out scenario

Synthetic best R^2: SNR 0.910, RMS 0.861, peak 0.825, crest 0.758, decay 0.622. Synthetic weak examples: dominant frequency -0.019, envelope-peak time 0.016.

C. Forward-Surrogate Results

The synthetic surrogate stopped at epoch 27. Table VI shows that amplitude, decay, and SNR-related features are learnable under domain randomization, while dominant frequency and envelope timing are poorly captured. Mean feature $R^2 = 0.409$ is useful as an engineering diagnostic but not sufficient justification to replace the physics signature in safety-relevant planning.

The measured LMSD scenario-held-out result is much weaker: mean $R^2 = 0.001$ and standardized MAE 0.449. High-band energy alone achieved $R^2 = 0.875$; many other features were negative out of sample. The adapter therefore demonstrates an honest data path and exposes domain shift rather than validating physical generalization. In particular, test Smooth L1 0.190 should not be interpreted as better than the synthetic 0.225 because the standardized targets and held-out units differ.

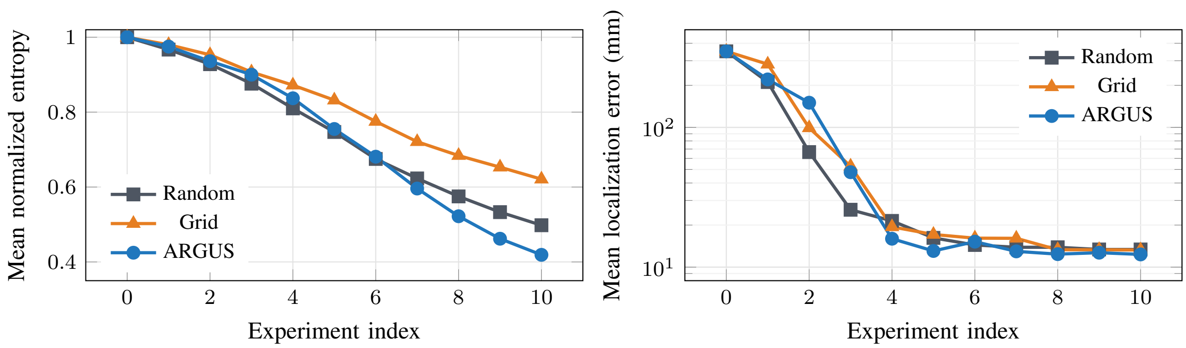

Fig. 3. Mean uncertainty (left) and localization-error (right, logarithmic axis) trajectories over the same 30 hidden defects. Lower is better.

D. Software Verification

The executed verification suite contains 41 passing backend tests. Besides the legacy deterministic simulation, posterior, planning, signal, API, persistence, surrogate, and data-contract tests, it covers dual-control switching, silence evidence, likelihood tempering, constraints, waveform bounds, discrepancy/OOD behavior, confidence gating, ledger tamper detection, bundle import, truth-sealed NPZ/CSV/WAV replay, additive migration, paired mismatch, injected faults, WebSockets, normalized healthy/damage/unknown state, accumulated sensor unreliability, duplicate-payload rejection, persisted emergency stop, and database hydration of assurance state. Three frontend utility tests pass; lint and the production build pass. The only observed backend warning is an upstream HTTP-test-client deprecation notice, not a runtime failure.

XIII. Why the Design Is Novel and Useful

The system-level research hypothesis is that hidden-state inference becomes more reliable and measurement-efficient when the controller may improve both the object estimate and the instrument estimate. Seven aspects distinguish the implemented prototype:

1) **Joint actionable uncertainty:** structural and metrology marginals determine the next action, rather than being passive confidence displays.
2) **Dual control:** the same policy selects calibration, diagnosis, exploration, or verification using explicit switching value.
3) **Counterfactual physical co-design:** geometry, frequency, amplitude, duration, waveform code, and fidelity are scored by the observation distributions expected under rival hypotheses.
4) **Risk and trust gating:** false decisions, another measurement, physical movement, excitation, time, discrepancy, and OOD evidence enter the stop/continue/abstain decision.
5) **Constraint-aware execution:** infeasible frequency, amplitude, spacing, no-go, and unavailable-device actions are rejected and explained before execution.
6) **End-to-end auditability:** decomposed alternatives, deterministic explanations, raw-response checksums, posterior evolution, model identity, and decisions form a verifiable evidence chain.
7) **Modality boundary:** device and inference interfaces permit future thermal, impedance, RF, optical, or tactile adapters without representing those modalities as already validated.

The potentially protectable technical nucleus, subject to legal review, is not "AI for defect detection," a generic digital twin, or information gain in isolation. A narrower candidate nucleus is the concrete sequence that (i) maintains coupled structural and measurement-system uncertainty, (ii) decides whether to calibrate the instrument or discriminate structural rivals, (iii) counterfactually co-designs an executable source–receiver waveform under risk, discrepancy, and safety constraints, (iv) gates the update and stopping decision with model-trust/OOD evidence, and (v) binds the observation, rationale, posterior transition, and model identity into a replayable integrity chain. Prior art may disclose some or all elements or render the combination obvious; only a professional search and claim chart can determine novelty, inventive step, eligible subject matter, and freedom to operate.

XIV. Limitations and Threats to Validity

A. Model and Inference Limits

The simulator is nondispersive and uses a single effective velocity. A real composite plate supports multiple dispersive, anisotropic modes and boundary reflections. The public structural marginal remains a one-dominant-anomaly location grid; size, severity, and type support in the joint interface is compact and not a validated multi-defect reconstruction. Baseline subtraction assumes the nominal digital twin is sufficiently calibrated. Multiplying successive likelihoods assumes conditional independence after approximate nuisance correction, while shared error can still make confidence over-sharp.

Confidence in Eq. (21) is a heuristic mixture, not a calibrated posterior coverage probability. NEO implements calibration diagnostics, discrepancy variables, likelihood tempering, OOD detection, and abstention, but the quick ECE of 0.715 demonstrates that implementation is not validation. The planner utilities and losses were engineered rather than elicited from a certified maintenance process. The conformal tail is withheld before ten scores and is only meaningful if future calibration scores are exchangeable with deployment. The Gaussian nuisance marginals, compact counterfactual features, online ridge correction, and short-horizon rollout are deliberate approximations.

B. Evaluation Limits

Thirty baseline synthetic cases provide a useful paired study, not population-scale validation. The baseline mean-error intervals cross zero; no claim of superior physical localization is warranted. The expanded NEO matrix has only two cases and is exclusively a software/research smoke test. Baselines do not include every advanced tomography, Gaussian-process design, reinforcement-learning, or optimal-array method. Hyperparameters were not evaluated in a nested holdout, and most studies share simulator assumptions with inference.

The LMSD experiment has only six independent damage scenarios, several with multiple masses. Hammer/accelerometer FRFs at 0–1600 Hz differ from the prototype's 1.2–6.2 kHz chirp interrogation. Added masses are scattering surrogates, not cavities or delaminations. Scenario-held-out performance correctly reveals that correlated channel count cannot replace independent physical diversity.

C. Operational Limits

Microphone measurements are vulnerable to airborne sound and ambient noise. Probe remounting changes coupling. Camera homography assumes a planar stationary panel and accurate clicks; inertial/visual metadata are quality proxies, not traceable metrology. Clock-skew checks, heartbeats, and reconnect logic do not provide hard real-time synchronization. ESP32 analog acquisition needs safe conditioning and synchronized excitation. The prototype is not certified NDE equipment and must not be used as the sole basis for safety-critical maintenance decisions.

XV. Physical Validation and Data-Collection Plan

A credible next study should use at least three nominally identical panels, eight repeatably mounted perimeter transducers, an instrumented exciter, five repeats per condition, and healthy measurements before and after each damage block. Reversible bonded masses, damping patches, or paired magnets are appropriate for commissioning. True cavities or delaminations require manufactured specimens—for example flat-bottom holes or embedded release films—and independent truth from ultrasound, radiography, or destructive sectioning where justified.

For the current acquisition contract, collect all 56 ordered non-self source–receiver pairs, three frequency bands, and five repeats. One physical state yields $56 \times 3 \times 5 = 840$ waveforms. Twenty known locations plus healthy references yield approximately 17,000 signals. Preserve raw waveforms and record specimen, material, thickness, defect geometry and truth method, source and receiver coordinates, excitation, repeat, temperature, humidity, coupling force, remount identifier, calibration, timestamp, and code version.

Split development data by complete defect location, stronger validation by complete specimen, and the final test by a blind panel whose truth remains hidden until predictions and stopping decisions are frozen. Offline planner evaluation additionally needs a complete counterfactual bank: every admissible experiment must be recorded for every hidden state so all policies query the same physical response table. Report error versus measurement budget, entropy calibration, success radii, abstention, remount robustness, and temperature shift.

XVI. Reproducibility

The repository provides commands for the complete workflow:

```
python run_argus.py
python run_edge_node.py --help
python scripts/run_neo_demo.py
  model_mismatch
python scripts/neo_benchmark.py --cases 2
python scripts/neo_benchmark.py --ablation
python scripts/neo_calibration.py --cases 4
python scripts/generate_counterfactual_bank.py
  tiny
python scripts/active_learn_surrogate.py
  --cases 180
python scripts/demo_simulation.py
python scripts/evaluate_model.py
python scripts/generate_dataset.py
python scripts/train_model.py
python scripts/download_lmsd_dataset.py
python scripts/prepare_lmsd_dataset.py
pytest
cd frontend
npm test
npm run lint
npm run build
```

The externally downloaded data are intentionally excluded from version control; DOI, license, source filenames, and checksums are preserved. Trained metadata records seeds, split units, sample counts, feature names, per-feature MAE, and R^2. Benchmark JSON preserves per-case trajectories and summary values. Evidence exports independently preserve the effective configuration, versions, ledger chain, planner alternatives, and response checksums. Every result in the NEO diagnostic table is labeled simulated unless a measured source is explicitly identified.

XVII. Conclusion

ARGUS NEO demonstrates how an NDE research prototype can be organized around the next experiment and the reliability of the measuring process, rather than only the next prediction. The implemented loop jointly tracks structural and metrology uncertainty; chooses calibration, diagnosis, exploration, or verification; co-designs bounded geometry and waveform; selects model fidelity; acquires or replays a response; tempers the update using quality, discrepancy, sensor reliability, and OOD evidence; explains alternatives; and records a verifiable chain before stopping or abstaining. ARGUS-X makes healthy and unsupported states explicit, refuses duplicate or infeasible evidence, exposes spatial rivalry without claiming confirmed multi-defect inference, and preserves a human-controlled emergency latch. The earlier matched-model baseline supports faster uncertainty concentration versus simple policies, while localization-error superiority remains unproven.

The quick NEO studies verify mechanisms and expose poor small-sample calibration, not inspection efficacy. The decisive milestone is a preregistered, specimen-held-out, blind physical campaign with a full counterfactual measurement bank, independent truth, calibrated coverage, and predefined decision losses. Until then ARGUS NEO is a capable, falsifiable research instrument—not certified inspection equipment.

Acknowledgment

The authors acknowledge the LMSD group at KU Leuven for releasing the LMSD 2021 plate-damage dataset under CC BY 4.0. Dataset use in ARGUS retains the published DOI and provenance.

Appendix A
Default Configuration

Table VII
Important Reproducible Defaults

Subsystem	Parameter	Value
Panel	width $\times$ height	0.60×0.40 m
Material	velocity	180 m/s
Material	attenuation	1.65 m^{-1}
Material	resonance	3100 Hz
Material	damping	95 s^{-1}
Material	noise standard deviation	0.007
Material	system delay	0.8 ms
Signal	sample rate	16 kHz
Signal	record duration	0.12 s
Belief	grid; base temperature	20×20; 4.8
Stop	confidence; normalized entropy	0.72; 0.38
Stop	maximum experiments	12
Planner	candidate count; top hypotheses	48; 24
Planner	I, D, C, R weights	1, .35, .16, .22
Planner	coverage weight	0.22
NEO	risk; calibration; trust weights	.55; .65; .30
NEO	horizon; beam; rollout samples	1; 4; 6
NEO	calibration threshold; switch margin	.62; .05
OOD	caution; abstain thresholds	.55; .82
Safety	amplitude; frequency; duration	.75; 7 kHz; .35 s
Evidence	minimum evidence weight	.08
Easy	radii; severity; noise multiplier	.10–.15; .78–.95; .55
Medium	radii; severity; noise multiplier	.065–.11; .58–.82; 1.0
Hard	radii; severity; noise multiplier	.04–.075; .38–.64; 1.55
Reproducibility	default seed	7

Appendix B
Interpretation Guide for Demonstrations

An entropy decrease means probability mass has concentrated, not necessarily that the mode is correct. Confidence combines concentration and local mass and is OOD-capped, but it is not a certified probability of correctness. Ground truth must remain hidden until the controller stops. Reveal physical error using panel dimensions, not normalized distance. If entropy falls while revealed error remains high, diagnose mismatch or overconfidence rather than calling the run successful. Never describe a two-case matrix, one controlled mismatch case, or simulated calibration coverage as general performance.

The strongest concise description for a judge is: "ARGUS NEO is an AI-guided scientific instrument that decides whether to calibrate itself or which safe physical measurement to take next. It predicts how rival defect hypotheses would respond to different probe geometries and waveforms, trades decision risk against time, movement, energy, and model trust, then explains and cryptographically records each update." The strongest defensible evidence statement is: "The 41-test implementation is complete and the earlier 30-case simulator baseline concentrated uncertainty more consistently than random or grid probing; the new NEO studies expose both a promising controlled mismatch mechanism and the need for blind physical calibration."

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